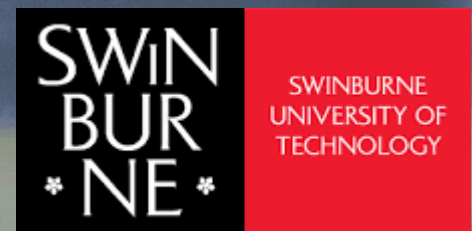




COS20028 – BIG DATA ARCHITECTURE AND APPLICATION LAB WEEK 10

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Week 10



**Analysing
Disparate
Data Sets
with Pig**



Lab Task 6



ANALYSING DISPARATE DATA SETS WITH PIG

Summary

In this exercise, you will practice combining, joining, and analysing the product sales data previously exported from Dualcore's MySQL database so you can observe the effects that the recent advertising campaign has had on sales.

Since this exercise is built on the achievements from previous labs, it is important that you successfully complete the previous lab exercises before starting this one.

Before we proceed with more sophisticated analysis, you should first calculate the number of orders Dualcore received each month for the three months before their ad campaign began (Feb. – Apr., 2013), as well as for the month during which their campaign ran (May, 2013).

Step 1: Show Per-Month Sales before and after Campaign

Change to the directory for this lab.

```
training@192-168-1-109:~/training_materials/analyst/exercises/disparate _ □ ×  
File Edit View Search Terminal Help  
[training@192-168-1-109 ~]$ cd /home/training/training_materials/analyst/exercis  
es/disparate_datasets/  
[training@192-168-1-109 disparate_datasets]$ █
```

Open the *count_orders_by_period.pig* file in your editor.

```
training@192-168-1-109:~/training_materials/analyst/exercises/disparate _ □ ×  
File Edit View Search Terminal Help  
[training@192-168-1-109 disparate_datasets]$ gedit count_orders_by_period.pig &  
[training@192-168-1-109 disparate_datasets]$ █
```

Modify the Code

Following the *FILTER* statement, create a new relation with just one field: the order's year and month.

- Hint: Use the *SUBSTRING* built-in function to extract the first part of the *order_dtm* field, which contains the month and year.

Count the number of orders in each of the months you extracted above.

Display the count by month to the screen.

```
(02,76170)
(03,84549)
(04,87853)
(05,115038)
```


Modify the Code and Execute It

- Once you have made the changes, try running the script against the data in HDFS.

```
*count_orders_by_period.pig X
data = LOAD '/dualcore/orders/part*' AS (order_id:int, cust_id:int, order_dtm:chararray);
--data = LOAD './orders/part*' AS (order_id:int, cust_id:int, order_dtm:chararray);

/*
 * Include only records where the 'order_dtm' field matches
 * the regular expression pattern:
 *
 * ^      = beginning of string
 * 2013   = literal value '2013'
 * 0[2345] = 0 followed by 2, 3, 4, or 5
 * -      = a literal character '-'
 * \\d{2} = exactly two digits
 * \\s    = a single whitespace character
 * .*    = any number of any characters
 * $      = end of string
 *
 * If you are not familiar with regular expressions and would
 * like to know more about them, see the Regular Expression
 * Reference at the end of the Exercise Manual.
 */
recent = FILTER data by order_dtm matches '^2013-0[2345]-\\d{2}\\s.*$';

-- TODO (A): Create a new relation with just the order's year and month
year_month = FOREACH recent GENERATE SUBSTRING(order_dtm,0,4) AS year_value:chararray,
                                     SUBSTRING(order_dtm,5,7) AS month_value:chararray;

-- TODO (B): Count the number of orders in each month
month_group = GROUP year_month BY month_value;
month_count = FOREACH month_group GENERATE group AS g_month:chararray, COUNT(year_month) AS order_count:int;

-- TODO (C): Display the count by month to the screen.
DUMP month_count;
```

```
[training@192-168-1-109 disparate_datasets]$ pig count_orders_by_period.pig
2020-09-28 05:17:46,126 INFO org.apache.pig.Main: Apache Pig version 0.10.0-cdh4
.2.1 (rexported) compiled Apr 22 2013, 12:04:54
2020-09-28 05:17:46,127 INFO org.apache.pig.Main: Logging error messages to: /ho
me/training/training_materials/analyst/exercises/disperate_datasets/pig_16012846
66117.log
(02,76170)
(03,84549)
(04,87853)
(05,115038)
[training@192-168-1-109 disparate_datasets]$
```

Step 2: Count Advertised Product Sales by Month

Our analysis from the previous step suggests that sales increased dramatically the same month Dualcore began advertising.

Next, you'll compare the sales of the specific product Dualcore advertised (product ID #1274348) during the same period to see whether the increase in sales was actually related to their campaign.

You will be joining two data sets during this portion of the task. Since this is the first join you have done with Pig, now it is a good time to mention a tip that can have a profound effect on the performance of your script.

Filtering out unwanted data from each relation *before* you join them, as we have done in our example, means that your script will need to process less data and will finish more quickly.

We will discuss several more Pig performance tips later in class, but this one is worth learning now.

Modify the Code

Edit the *count_table_orders_by_period.pig* file and implement the following changes.

```
[training@192-168-1-109 disparate_datasets]$ gedit count_table_orders_by_period.pig &
```

Join the two relations on the *order_id* field they have in common.

Create a new relation from the joined data that contains a single field: the order's year and month, similar to what you did previously in the *count_orders_by_period.pig* file.

Group the records by month and then count the records in each group.

Display the result to your screen.

Modify the Code and Execute It

- Once you have made the changes, try running the script against the data in HDFS.

```
count_orders_by_period.pig x count_tablet_orders_by_period.pig x
orders = LOAD '/dualcore/orders/part*' AS (order_id:int, cust_id:int, order_dtm:chararray);

details = LOAD '/dualcore/order_details/part*' AS (order_id:int, prod_id:int);

-- include only the months we want to analyze
recent = FILTER orders BY order_dtm matches '^2013-0[2345]-.*$';
--DESCRIBE recent;

-- include only the product we're advertising
tables = FILTER details BY prod_id == 1274348;
--DESCRIBE tables;
--DUMP tables;

-- TODO (A): Join the two relations on the order_id field
joined = JOIN recent BY order_id, tables BY order_id;
--DESCRIBE joined;
--DUMP joined;

-- TODO (B): Create a new relation containing just the month
just_month = FOREACH joined GENERATE SUBSTRING(recent::order_dtm,5,7) AS order_month:chararray;
--DUMP just_month;
--DESCRIBE just_month;

-- TODO (C): Group by month and then count the records in each group
grouped = GROUP just_month BY order_month;
counted = FOREACH grouped GENERATE group, COUNT(just_month) AS month_sale:int;

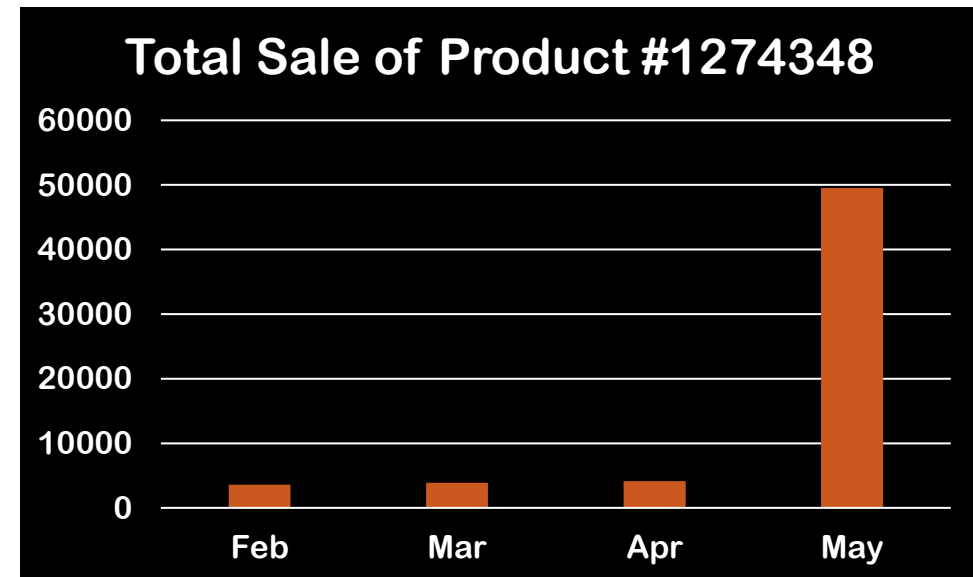
-- TODO (D): Display the results to the screen
DUMP counted;
```

```
training@192-168-1-109:~/training_materials/analyst/exercises/disparate _ x
File Edit View Search Terminal Help
[training@192-168-1-109 disparate_datasets]$ pig count_tablet_orders_by_period.p
ig
```

Discussion

- If we look at the particular product ID, you can see the spike in May of the total sale.
- Since this is contributed by a single product, which was put on the advertisement, we can conclude that the advertisement campaign works.

```
training@192-168-1-109:~/training_materials/analyst/exercises/disparate _ □ x
File Edit View Search Terminal Help
[training@192-168-1-109 disparate_datasets]$ rm -rf *.log
[training@192-168-1-109 disparate_datasets]$
[training@192-168-1-109 disparate_datasets]$ pig count_tablet_orders_by_period.pig
2020-09-28 07:44:16,302 INFO org.apache.pig.Main: Apache Pig version 0.10.0-cdh4
.2.1 (rexported) compiled Apr 22 2013, 12:04:54
2020-09-28 07:44:16,303 INFO org.apache.pig.Main: Logging error messages to: /home/training/training_materials/analyst/exercises/disparate_datasets/pig_1601293456294.log
(02,3598)
(03,3904)
(04,4134)
(05,49514)
[training@192-168-1-109 disparate_datasets]$
```



Discussion

- Only look at a single product is not the best way to present the result.
- You can further look at the stacked up result to find the ratio between the item put on the advertisement against all other products in the sale pool.

